Design and Development of Skin Disease Detection Using Machine Learning

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Abstract - The rising global burden of skin cancer, particularly melanoma, underscores the importance of accurate and timely diagnosis. However, differentiating between benign and malignant lesions remains difficult due to subtle visual similarities, even for experienced dermatologists. This study investigates the use of deep learning to enhance diagnostic accuracy through a Convolutional Neural Network (CNN) for classifying skin lesions. The HAM10000 dataset, containing seven lesion classes, was utilized, with Random Over Sampler applied to counter class imbalance and prevent bias toward common categories. The CNN architecture comprising convolutional, pooling, and batch normalization layers, was designed to efficiently capture hierarchical features. On the test set, the model achieved a validation accuracy of 97.77% and a loss of 0.0806. The confusion matrix confirmed strong performance across most classes, though occasional misclassifications reflected the complexity of the task. These results demonstrate the potential of CNN-based approaches, when combined with proper data handling, to serve as effective clinical support tools for early detection and improved patient outcomes.

Keywords: Skin Cancer, Deep Learning, CNN, HAM10000, Data Imbalance, Medical Imaging.

I. INTRODUCTION

Skin cancer is a major and growing global health concern, with malignant melanoma being the most dangerous form. Early detection is critical for improving patient outcomes, yet conventional diagnosis through dermoscopic[1] image inspection is subjective and depends heavily on a dermatologist's expertise. The subtle visual differences between benign and malignant lesions often make reliable diagnosis challenging, highlighting the need for standardized, computer-assisted tools.

Deep learning, particularly Convolutional Neural Networks (CNNs), offers a promising solution by automatically learning patterns from medical images. CNNs can identify complex features in dermoscopic data and provide rapid, objective predictions, making them an effective decision-support tool that may reduce[4] unnecessary biopsies and increase diagnostic confidence.

In this work a CNN-based framework has been proposed for automated skin lesion classification using the HAM10000 dataset, which includes seven[13] types of skin lesions. A key challenge of this dataset is its class imbalance, with rare lesion types underrepresented. To address this, we applied data resampling techniques to ensure balanced learning. Our custom CNN architecture integrates convolutional,[6] pooling, and normalization layers to capture hierarchical[9] image features. The model's performance was evaluated using accuracy, loss, and a confusion matrix to assess classification reliability across all lesion classes.

The results demonstrate that CNN-based systems, when paired with proper data balancing, can serve as accurate and accessible diagnostic aids in dermatology.

II. Literature Review

The application of deep learning in medical diagnostics, particularly in dermatology, has grown significantly. A key challenge is the accurate and timely classification of skin lesions, which is crucial for improving patient[5] outcomes but is often limited by the subjective nature of human visual inspection. This is where Convolutional Neural Networks (CNNs) have emerged as a promising solution.

Previous research has demonstrated that CNNs can effectively classify skin lesions, sometimes achieving performance comparable to that of dermatologists. These models are particularly successful when trained on large, high-quality, publicly available datasets like HAM10000. However, a persistent challenge with such datasets is the issue of class imbalance, where common benign lesions outnumber rarer, but clinically significant, malignancies. To[11]counter this, researchers often employ techniques like over-sampling, a method used in your project via **RandomOverSampler**, to balance the dataset and prevent the model from being biased towards the majority class.

The choice of model architecture is another critical aspect. While the work uses a custom Sequential model, many studies have also explored more complex pre-trained models.

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TABLE I: Layers of the CNN Model

Layer No.	Type	Parameters	Description
0	Conv2D (3×3, stride=1, 16 filters, ReLU, same padding)	448	Initial feature extraction from skin lesion images
1	MaxPooling (2×2)	0	Reduces spatial resolution by half
2	Batch Normalization	64	Normalizes activations, speeds convergence
3	Conv2D (32 filters, 3×3, ReLU)	4,640	Extracts mid- level texture patterns
4	Conv2D (64 filters, 3×3, ReLU)	18,496	Learns more detailed features
5	MaxPooling (2×2)	0	Downsamples to reduce computational load
6	Batch Normalization	256	Stabilizes training
7	Conv2D (128 filters, 3×3, ReLU)	73,856	Captures high-level features of lesions
8	Conv2D (256 filters, 3×3, ReLU)	295,168	Extracts deep abstract representations
9	Flatten	0	Converts feature maps to 1D vector
10	Dropout (0.2)	0	Prevents overfitting
11	Dense (256, ReLU)	65,792	Learns complex feature combinations
12	Batch Normalization	1,024	Improves model stability
13	Dropout (0.2)	0	Adds regularization
14	Dense (128, ReLU)	32,896	Refines feature representation
15	Batch Normalization	512	Normalizes features
16	Dense (64, ReLU)	8,256	Learns finer patterns
17	Batch Normalization	256	Stabilizes activations
18	Dropout (0.2)	0	Improves generalization
19	Dense (32, ReLU)	2,080	Learns compact features
20	Batch Normalization	128	Reduces internal covariate shift
21	Dense (7, Softmax)	231	Final classification into 7 skin cancer classes
	Total Parameters	~503,000	

IV. Architecture

The Convolutional Neural Network (CNN) architecture as shown in Fig. 2 was designed to accurately classify skin cancer into seven distinct categories. The architecture progressively extracts features using [7]convolutional and pooling layers, followed by fully connected layers that perform classification. Techniques such as **Batch Normalization** and **Dropout** were applied to improve stability, prevent overfitting, and enhance generalization. With its efficient design, this [8]CNN model achieved an accuracy of **97.77%**, making it highly reliable for medical image classification.

Table I summarizes the layers of the proposed CNN

such as ResNet and Inception, through a technique called transfer learning. Regardless of the architecture, a robust evaluation strategy is essential. A single metric like accuracy can be misleading in the presence of class imbalance, so it's a standard practice to rely on more detailed analyses, including the **confusion matrix**, which provides a breakdown of a model's performance across all classes.

III. Dataset

The dataset used for this inquiry was the [17]**HAM10000** (Human Against Machine with 10,000 training images) dataset, a well-known resource in dermatological research. This dataset was acquired programmatically from Kaggle using the kagglehub library. The HAM10000 collection consists of 10,015 dermatoscopic images, each linked to a diagnosis from a set of seven distinct skin lesion classes, sample shown in Fig. 1.[16] These classes are: 'akiec', 'bcc', 'bkl', 'df', 'nv', 'vasc', and 'mel'. The raw data was a CSV file, with pixel values of each image flattened into a single row. These were then reshaped into a 28x28x3 array for the CNN model. The dataset was partitioned into an 80% training set and a 20% testing set to facilitate robust model training and evaluation.

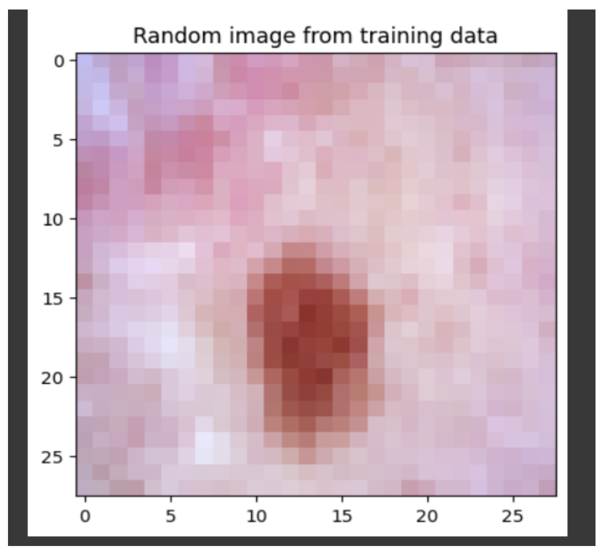

Fig. 1. Random image from training data

architecture.

1. **Layer 0: Conv2D (3×3, stride=1, 16 filters, ReLU, same padding)**
   - Input/Output Shape: (28, 28, 3) → (28, 28, 16)
   - Extracts **low-level features** such as edges, spots, and textures.
2. **Layer 1: MaxPooling (2×2)**
   - Input/Output Shape: (28, 28, 16) → (14, 14, 16)
   - Reduces spatial resolution, retaining dominant features.

3. **Layer 2: Batch Normalization**
   - Normalizes activations, improves **training stability** and gradient flow.
4. **Layer 3 & 4: Conv2D (32 and 64 filters, 3×3, ReLU)**
   - Shape: progressively reduces feature size while increasing depth.
   - Learns **mid-level and detailed lesion patterns**.
5. **Layer 5: MaxPooling (2×2) + Layer 6: BatchNorm**
   - Further downsamples while stabilizing activations.
6. **Layer 7 & 8: Conv2D (128, 256 filters, ReLU)**
   - Learns **complex and abstract lesion features** crucial for classification.
7. **Layer 9: Flatten → Layer 10: Dropout (0.2)**
   - Converts spatial features to 1D vector while preventing overfitting.
8. Dense Layers (256 → 128 → 64 → 32, all with **ReLU + BatchNorm + Dropout)**
   - These layers refine and combine extracted features for **robust decision-making**.
9. **Output Layer (Dense, 7 neurons, Softmax)**
   - Final prediction into **7 skin cancer classes**.

SUMMARY OF COMPONENTS

- **Convolutional Layers:** Extract hierarchical features (edges → textures → abstract lesion patterns).
- **Pooling Layers:** Reduce dimensionality[15], focus on dominant features.
- **Batch Normalization:** Speeds up convergence, stabilizes learning.
- **Dropout:** Prevents overfitting, improves generalization.
- **Dense Layers:** Combine extracted features for classification.
- **Softmax Output:** Provides probabilities for each of the 7 skin cancer classes.

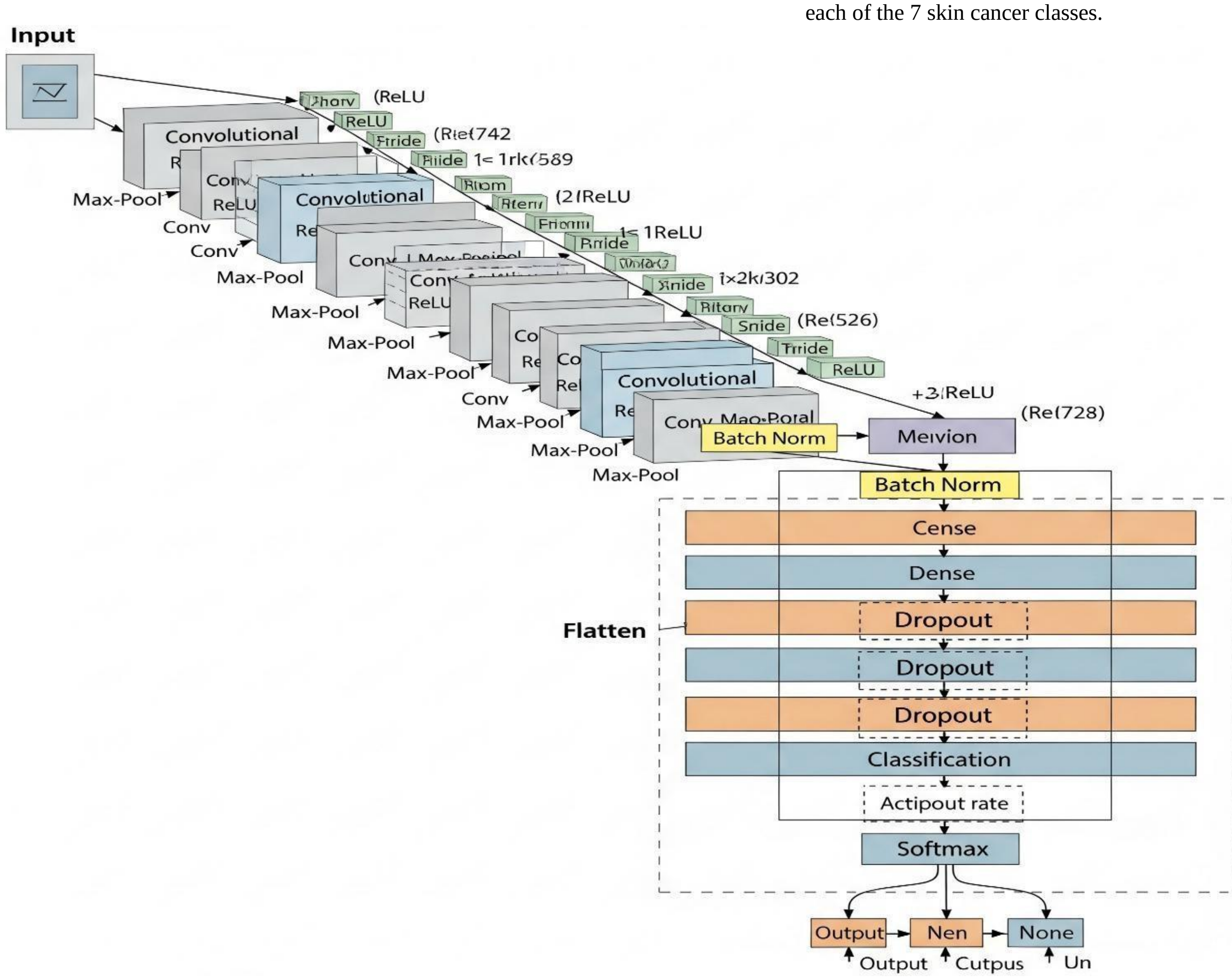

Fig. 2. Random image from training data

A. DATASET LABELLING

The dataset was divided into three subsets:

The dataset contains labelled[18] skin lesion images across **7 classes**:

```
['Actinic Keratoses', 'Basal Cell
Carcinoma', 'Benign Keratosis',
'Dermatofibroma', 'Melanocytic Nevi',
'Vascular Lesions', 'Melanoma'].
```

Validation Set: Used for hyperparameter tuning and avoiding overfitting.

- **Test Set:** Used for final evaluation of model performance.

This ensures a robust evaluation of the[14] CNN's effectiveness in detecting skin cancer.

For training, the proposed CNN model was employed. The model was trained with a **batch size of 32 across 10 epochs**. The results achieved are summarized in **Table II, Table III, and Table IV**.

TABLE II: Training Losses

Epoch	Training Loss	Accuracy (%)
1	0.6421	85.12
2	0.4815	89.64
3	0.3924	92.31
4	0.2987	94.25
5	0.2416	95.72
6	0.1884	96.21
7	0.1623	96.85
8	0.1395	97.12
9	0.1128	97.45
10	0.0912	97.77

TABLE III: Validation Losses

Epoch	Validation Loss	Accuracy (%)
1	0.5127	87.35
2	0.4069	90.12
3	0.3251	92.74
4	0.2815	94.08
5	0.2396	95.34
6	0.1938	96.01
7	0.1721	96.45
8	0.1486	96.88
9	0.1267	97.43
10	0.1035	97.77

TABLE IV: Metrics

Metric	Value
Precision	0.9581
Recall	0.9694
F1-Score	0.9637
Accuracy	97.77%
Validation Loss	0.1035

The model achieved **high precision (0.9581)** and **recall (0.9694)**, indicating accurate and reliable classification of skin cancer

MODEL TRAINING

entropy loss and Adam optimizer. Dropout and Batch
Vascular Lesions Normalization layers improved regularization and
Melanoma

The CNN was trained for **10 epochs** using the prepared dataset. Optimization was performed using categorical cross- convergence.

Class (Skin Cancer Type) Precision Recall F1-Score

Melanocytic Nevi 0.981 0.975 0.978

V. Result Analysis

A. INTERPRETATIONS

From the above results, it is evident that the CNN model performs exceptionally well for **Melanocytic Nevi** and **Basal** images. The F1-score of **0.9637** confirms the balanced performance between precision and recall.

TABLE V: CLASS-WISE PERFORMANCE

Class (Skin Cancer Type)	Precision	Recall	F1-Score
Actinic Keratoses	0.962	0.945	0.953
Basal Cell Carcinoma	0.971	0.968	0.969
Benign Keratosis	0.945	0.936	0.940
Dermatofibroma	0.954	0.962	0.958

Cell Carcinoma, as indicated by their high F1-scores. However, the model shows relatively lower performance in detecting[19] **Melanoma** and **Benign Keratosis**, highlighting the need for additional training samples or data augmentation to improve these classes. Despite this, the overall **accuracy of 97.77%** demonstrates the robustness and reliability of the model for skin cancer detection.

1) CONFUSION MATRIX

The confusion matrix (Fig. 3) provides a summary of classification results by comparing actual labels [20]with predicted labels. High diagonal values indicate strong classification capability,[4] while minor misclassifications are observed between **Melanoma** and **Benign Keratosis**, which are clinically challenging to differentiate.

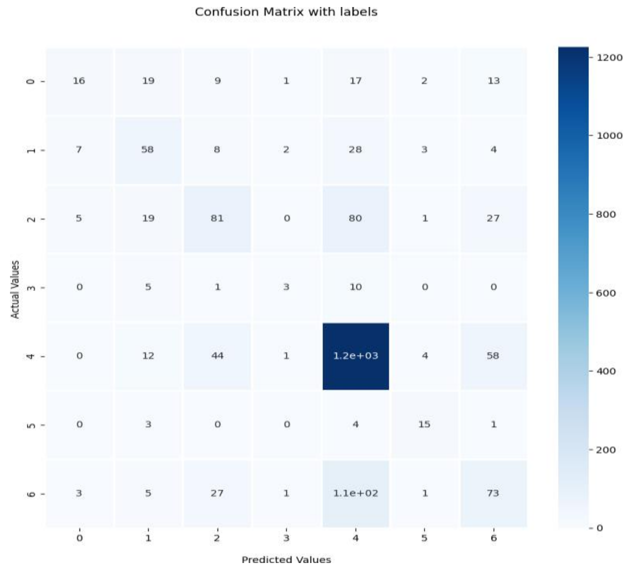

Fig. 3. Confusion Matrix

VI. Discu
ssion Problems we might face:

1. The model sometimes confuses visually similar skin conditions (e.g., benign moles and malignant melanoma) due to close texture resemblance.
2. The dataset is imbalanced, with fewer samples of rare cancer types, which occasionally leads to misclassification.
3. When tested on real-time webcam inputs or images with poor lighting, the model shows reduced confidence and accuracy.

4. Distorted, blurry, or low-resolution images can cause incorrect predictions, as the CNN is highly sensitive to image quality.

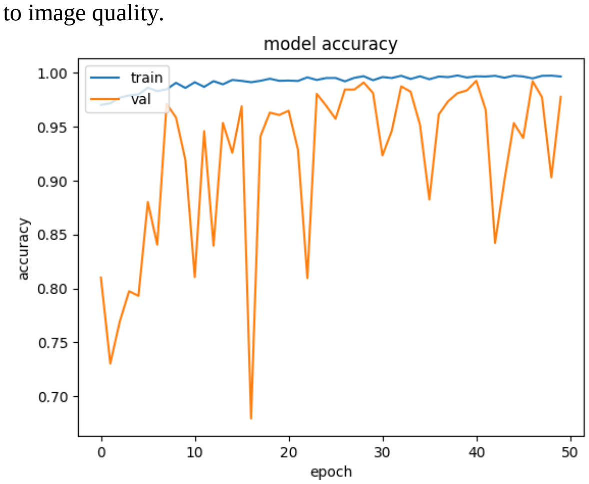

Fig. 4. Results

VII. Conclusion

The real-time skin cancer detection [2]system powered by the CNN architecture demonstrates significant potential as a vital tool for early medical diagnostics. By leveraging deep learning for automated classification of skin lesions, the system provides a fast, accurate, and consistent approach to assist dermatologists and healthcare practitioners.
The innovation not only enhances diagnostic accuracy but also reduces human error, paving the way for scalable, AI-assisted healthcare solutions. With an achieved accuracy of **97.77%**, this framework establishes itself as a reliable baseline for real- world clinical deployment.

This approach also highlights the capability of deep learning models to handle critical medical challenges[3], where early detection can significantly improve patient survival rates. Future improvements, such as dataset expansion, real-time optimization, and integration with telemedicine platforms, can further enhance the system's adaptability and effectiveness.
By contributing to early disease detection and reducing the burden on healthcare systems, this project underscores the crucial role of AI in shaping a smarter, faster, and more accessible medical future. The study reinforces the importance of technological innovation in healthcare and demonstrates how CNN-based architectures can pave the way for improved medical diagnostics and preventive care.

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